FLDCF: A Collaborative Framework for Forgery Localization and Detection in Satellite Imagery

Jialu Sui, Ding Ma, C.-C. Jay Kuo, *Life Fellow, IEEE*, and Man-On Pun, *Senior Member, IEEE*

***Abstract*— Satellite images are highly susceptible to forgery due to various editing techniques. Traditional forgery detection methods, designed for natural images, often fail when applied to satellite images because of differences in sensing technology and processing protocols. The rise of generative models, such as diffusion models, has further complicated the detection of forgeries in satellite images. This study tackles these challenges from both methodological and data perspectives. We introduce a multitask forgery localization and detection collaborative framework (FLDCF), comprising a multiview forgery localization network (M-FLnet) and a forgery detection network. The M-FLnet, leveraging a content-based prior, generates forgery masks that serve as auxiliary information to improve the detection network's accuracy. Conversely, the detection network refines these masks, reducing noise for authentic images. Furthermore, two novel forgery datasets, namely, Fake-Vaihingen and Fake-LoveDA, are derived from the Vaihingen and LoveDA satellite image sets, respectively, by exploiting the latest generative models. These datasets represent the first open-source datasets for forgery localization and detection in remote sensing. Extensive experimental results on Fake-Vaihingen and Fake-LoveDA demonstrate that the proposed FLDCF can effectively detect sophisticated forgeries in satellite imagery. The source code and datasets in this work are available at https://github.com/littlebeen/Forgery-localization-for-remote-sensing.**

***Index Terms*— Forgery dataset, forgery localization and detection, remote sensing.**

I. Introduction

SATELLITE imagery is widely used in fields such as disaster response, environmental monitoring, and urban planning. However, digital image processing tools have made these images susceptible to alterations and manipulations for economic gain and other malicious purposes. Tampered images can obscure buildings, disasters, and pollution, potentially leading to negative consequences through misinformation [1], [2]. Such forged satellite images can significantly impact downstream analyses, resulting in misguided decisions. In addition, advancements in deep learning (DL) techniques have simplified the creation of highly realistic forgery images [3]. As a result, the localization and detection of manipulated images have become pressing challenges that require urgent improvement.

Recently, significant progress has been made in image forgery tasks, which can be broadly categorized into forgery localization [4], [5] and forgery detection [6], [7]. While these two tasks are closely interconnected, they differ in their specific objectives: forgery localization identifies precise locations where alterations have been made, while forgery detection aims to verify the overall authenticity of an image. By integrating DL-based techniques, both forgery localization and detection [8], [9] achieve superior performance compared to traditional approaches in detecting various types of forgery manipulations. The creation of image forgery can be categorized into two approaches, namely, conventional forgery manipulation [4], [10], [11] and generative model-produced forgery manipulation [12], [13], [14]. In traditional digital image processing, conventional forgery manipulations include techniques, such as copy-paste, copy-move, and splicing, typically carried out using popular tools such as Photoshop. These methods often result in noticeable differences in local features between the original and manipulated areas. In contrast, advancements in generative models have introduced new forms of forgeries, such as inpainting and whole-image generation. These generative model-produced forgeries tend to be of higher quality, posing greater challenges for localization and detection compared to conventional manipulations [15].

Some DL forensic techniques [16] are tailored specifically for detecting conventional forgery manipulations based on distinct optical features between authentic and altered areas. For instance, Zhang et al. [17] introduced a two-stream network comprising a label mask stream and an edge-guided stream for splicing localization. This network integrated low-level feature maps from shallow layers with high-level feature maps from deeper layers to enhance the reliability of splicing localization. In the field of remote sensing, Yarlagadda et al. [18] developed a method to learn feature representations of pristine satellite images and trained a one-class support vector machine to identify their distribution. However, forgeries created through conventional methods often exhibit coarse characteristics that can be easily detected by both DL models and human perception. First, models designed solely around optical differences from conventional manipulation methods may struggle with detecting forgery images of high similarity. Second, as novel forgery generation methods advance and conventional manipulations become less common, the effectiveness of forgery localization and detection models designed specifically for conventional techniques is progressively limited.

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Driven by the rapid development of generative models [19], [20], [21], [22], high-quality forgery generation has become increasingly sophisticated, posing considerable challenges for forgery localization and detection tasks. Consequently, there is a pressing need for novel forgery localization and detection models capable of effectively handling those forgery images created by generative models [23]. In the domain of computer vision tasks, several models have been developed to address complex forgery manipulations. For instance, Dong et al. [24] proposed multiview feature learning, which leverages tampering boundary artifacts and noise views to learn semantic-agnostic features adaptable across various types of manipulations. However, those models designed for natural images may not be able to fully meet the requirements for forgery localization in remote sensing due to the significant differences in image acquisition methods and sensing technologies. In the remote sensing field, there is currently a lack of methods and datasets specifically designed to handle the high-quality forgeries generated by novel large generative models.

In principle, forgery localization and detection are closely related tasks. The detection score can be derived from the response of the predicted manipulation mask, assuming that at least one area of the forged image elicits a high response, while no area of the pristine image does. However, most prior works assume the existence of manipulation in all input images. As a result, this could cause many false alarms on pristine images and make the detection unreliable. Besides relying on the mask produced by forgery localization, there are many methods that only make forgery detection based on the image itself [6], [7], [25], yielding variable results depending on the characteristics of the image content and the specific features of the manipulation. This variability can lead to inconsistent performance across different datasets and types of manipulations.

To address the challenges of forgery localization [26] and detection [27], [28] in remote sensing images, this work introduces improvements in both methodology and dataset creation. Methodologically, we introduce the first multitask forgery localization and detection collaborative framework (FLDCF), which integrates these tasks synergistically for enhanced performance in identifying high-quality forged images. The collaborative framework uses forgery localization masks as additional guidance for forgery detection that ensures its accuracy while leveraging the outputs of forgery detection to rectify minor errors that occur in authentic image masks. Specifically designed for high-quality generated forgeries, a multiview forgery localization network (M-FLnet) equipped with a two-class classifier is proposed. This network comprehensively analyzes global image information to accurately delineate forgery areas based on multiview learning. M-FLnet incorporates a content-based prior to learn global features from authentic images and capture specific characteristics of generated forgeries. Furthermore, we develop two benchmark datasets generated from different satellite image sources by exploiting the latest generative models. To the best of our knowledge, these datasets are the first datasets specifically designed with generative models for forgery localization in remote sensing, posing a significant challenge to the forgery localization and detection tasks due to their high-quality generation. The main contributions of this work are summarized as follows.

1) We propose a novel multitask FLDCF that integrates forgery localization and detection tasks. This combined approach enhances the performance of each task synergistically. Furthermore, we introduce M-FLnet specifically designed to accurately outline forgery areas based on global image features derived from multiview learning.
2) A content-based prior is designed to guide forgery localization by leveraging global feature information from authentic locations and characteristics specific to forgery locations.
3) The first generative model-produced high-quality forgery benchmarks called Fake-Vaihingen and Fake-LoveDA are developed for evaluating forgery localization and detection methods in remote sensing. These datasets are generated using two large inpainting models based on the novel diffusion architecture.

The rest of this article is organized as follows. Section II provides an overview of existing forgery datasets and methods, whereas Section III outlines our proposed forgery datasets. After that, the proposed FLDCF is detailed in Section IV before experimental results and insights are explored in Section V. Finally, Section VI presents concluding remarks.

II. Related Work

A. DL for Forgery Generation and Forgery Localization and Detection

Recently, DL-based forensic techniques have surpassed traditional methods in performance, largely due to the adoption of advanced architectures such as convolutional neural networks (CNNs), generative adversarial networks (GANs), and diffusion models. These architectures play a pivotal role in both generating and detecting fake content. More specifically, CNN-based models utilize convolutional layers for effective feature extraction, which is crucial for tasks such as detecting and localizing forgeries [29], [30]. The integration of attention modules [31] into CNN architectures has further bolstered their ability to extract global features. Furthermore, GAN-based models leverage unique training strategies involving generators and discriminators, enhancing their capabilities in both generating and detecting forgeries. For instance, Chen et al. [6] applied StyleGAN [32] and LightGAN [33] in the realm of satellite image forgery generation. In the domain of forgery localization, recent work [34] has utilized homology-aware discriminators to significantly enhance localization accuracy. Finally, diffusion-based models [19], [35] have gained traction for their prowess as generative models, notably improving the quality of generated images.

B. Forgery Localization

Forgery localization generates a specialized mask to pinpoint all locations of tampering in an image. Existing

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TABLE I
COMPARISON BETWEEN EXISTING FORGERY DATASETS AND OUR GENERATIVE MODEL-PRODUCED DATASET

Dataset	Image Size	Number	Generated Way
[18]	650×650	130	Splicing
Deepfake [7]	256×256	4,023	CycleGAN
Geo-deepfake1 [6]	128×128	12,069	StyleGAN
Geo-deepfake2 [6]	128×128	12,069	LightGAN
Fake-Vaihingen (proposed)	256×256	2,624	Repaint, LaMa
Fake-LoveDA (proposed)	512×512	8,572	Repaint, LaMa

techniques can be categorized into conventional and advanced methods. Some focus on traditional tampering operations, such as [5], [36], while others [24], [37], [38] are tailored for handling more complex manipulations. As advanced high-performance generative models are increasingly used in creating forgeries, it becomes crucial to develop robust solutions capable of detecting and mitigating such manipulations. In computer vision, Zeng et al. [37] proposed a mask-guided query-based transformer framework (MGQFormer), which used ground-truth masks to guide the learnable query token in identifying the forged regions. In addition, Wu et al. [38] introduced a leading baseline image forgery detector with a training strategy focused on enhancing robustness against image distortions. However, in the field of remote sensing, there has been a notable absence of research focused on accurately localizing forgeries generated by advanced generative models to ensure the integrity and reliability of remote sensing data and analysis.

C. Forgery Image Detection

Forgery detection [7], [25] aims to provide a definitive assessment of the authenticity of an entire image. Addressing this challenge, Nguyen et al. [39] introduced a distinctive approach using capsule networks to detect counterfeit images across various forgery scenarios, encompassing both complete and partial alterations. In the realm of deepfake detection in remote sensing, Chen et al. [6] developed Geo-DefakeHop that identified the most discriminative channels by analyzing response differences between authentic and fake images across a filter bank. It utilized soft decision scores from these channels as features, selecting a subset to the ensemble for a final binary decision. However, the performance of these methods can vary depending on the content and manipulation of the forgery images, limiting their ability to comprehensively address forgery detection across all types of images and forgery.

D. Forgery Datasets

There is an urgent need for a high-quality benchmark in forgery localization and detection, given the advancements in generative models. Table I outlines several notable existing datasets used for forgery localization and detection.

One notable dataset, proposed in [18], focuses on satellite image forgery localization through splicing credible objects such as airplanes and clouds onto authentic images. However, because these inserted objects come from different sources, their optical characteristics differ significantly from satellite imagery, making them easily detectable by localization models and even human perception. Moreover, this dataset is not open source. In contrast, datasets introduced in [6] and [7] employ generative models such as CycleGAN [40], StyleGAN [32], and LightGAN [33] for generating forgery images. However, these models are outdated compared to the latest generative models such as diffusion-based models that produce much higher quality forgeries. Furthermore, these datasets focus on forgery detection on entire images, in lieu of identifying specific forgery areas, limiting their applicability in forgery localization tasks. Motivated by the discussions above, it is imperative to develop high-quality, generative model-produced datasets in remote sensing specifically tailored for forgery localization and detection, which will bridge the gap between high-quality forgery images created by advanced generative models and existing coarse manipulated datasets.

TABLE II
NUMBER OF IMAGES IN TRAINING AND TEST SETS

Dataset	Set	Authentic Image	Repaint	LaMa
Fake-Vaihingen	training	555	752	792
	test	139	188	198
Fake-LoveDA	training	1724	2564	2564
	test	428	648	644

III. GENERATIVE MODEL-PRODUCED FORGERY DATASETS

To assess the efficacy of fake localization and detection in remote sensing, we introduce the first generative forgery benchmarks utilizing two high-quality satellite datasets: Vaihingen [41] and LoveDA [42]. These benchmarks, named Fake-Vaihingen and Fake-LoveDA, aim to advance the capabilities of forgery localization and detection. As mentioned in Section II-D, existing datasets for remote sensing forgery localization rely on conventional methods such as copy-paste and splicing, which produce lower quality, easily detectable forged images. More critically, these datasets are not open source. Although there are forgery detection datasets based on generative models such as CycleGAN and StyleGAN, their performance is noticeably inferior compared to our diffusion-based large generative models.

As shown in Table II, two pretrained large-scale inpainting models are employed for each dataset, namely, Repaint [21] and LaMa [22]. These generative models have been trained extensively on diverse datasets to ensure the generation of high-quality forgeries. The forgery process involves applying masks with random shapes and positions, ranging from 10% to 40% coverage. These masks include both regular rectangles and irregular shapes, further challenging the forgery localization and detection tasks. Fake-Vaihingen and Fake-LoveDA are designed to serve as robust benchmarks that challenge and enhance the capabilities of forgery localization and detection in remote sensing applications.

A. Fake-Vaihingen

Fake-Vaihingen is derived from the Vaihingen dataset, which is widely used for segmentation tasks [43], [44], [45]

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and features a spatial resolution of 9 cm. The original dataset includes 33 large images labeled with six classes: "roads," "buildings," "cars," "low vegetation," "trees," and "clutter." For Fake-Vaihingen, we select 11 and 12 images separately from this dataset to undergo manipulation using Repaint and LaMa inpainting models, respectively, applying random masks. The remaining images are kept unaltered to serve as authentic images for forgery detection experiments. Due to computational constraints, the large images are cropped into patches of size 256 × 256 pixels. The specific numbers of images generated by Repaint and LaMa are detailed in Table II. A training-to-testing set ratio of 8 : 2 is utilized for subsequent experiments. The establishment of Fake-Vaihingen, thus, provides a controlled environment to evaluate the performance of forgery localization and detection algorithms under realistic conditions, leveraging high-resolution satellite imagery and manipulated patches.

In addition to Fake-Vaihingen, we also propose Splice-Vaihingen on Vaihingen to assess the image quality resulting from various manipulation methods and evaluate the performance of each localization model. The setup for Splice-Vaihingen follows that of Fake-Vaihingen in all aspects except for the method used to generate forgery images. Instead of using the Repaint and LaMa, Splice-Vaihingen employs a splicing technique to create forged images. In our study, we specifically compare the quality of the forgery images and the accuracy of localization results between the splicing dataset and the generative model-produced dataset. This comparison aims to underscore the necessity and effectiveness of using generative model-produced benchmarks for evaluating forgery localization and detection methods in remote sensing applications. Detailed discussions are provided in Sections III-C and V-G.

B. Fake-LoveDA

Fake-LoveDA is constructed from the LoveDA dataset, a well-known segmentation dataset with a spatial resolution of 30 cm. The original dataset includes 5987 images labeled into seven classes: "background," "road," "buildings," "water," "barren," "forest," and "agriculture," each with an image size of 1024 × 1024 pixels. Methods for generating masks applied in Fake-LoveDA during the forgery process are similar to those in Fake-Vaihingen. The training-to-testing set ratio is consistent with Fake-Vaihingen, set at 8:2 for both datasets. Specific details regarding the number of images in each dataset can be found in Table II.

C. Data Analysis

To assess the quality of forgery in those two forgery datasets, we utilize several unsupervised image quality metrics: NIQE [46], BRISQUE [47], MUSIQ [48], and DBCNN [49]. These metrics are applied to evaluate both the forgery images and their corresponding authentic images, with results summarized in Table III. Since those metrics are no-reference image quality assessments primarily developed for natural images, we consider datasets with similar metric values to be comparable in quality, rather than focusing on the exact values of each individual metric.

TABLE III

FORGERY IMAGE QUALITY EVALUATION. ↑ AND ↓ REPRESENT HIGHER BETTER AND LOWER BETTER, RESPECTIVELY

Dataset	Mask(%)	NIQE↓	BRISQUE↓	MUSIQ↑	DBCNN↑
Splice-Vaihingen	22.66	4.798	12.739	3.740	0.530
Fake-Vaihingen	22.66	5.223	12.204	3.489	0.455
Vaihingen	/	5.595	13.010	3.492	0.461
Fake-LoveDA	17.64	3.913	27.571	3.924	0.324
LoveDA	/	4.055	28.241	3.929	0.330

TABLE IV

FORGERY IMAGE QUALITY EVALUATION OF REPAINT AND LAMA WITH THE SAME MASK ON VAIHINGEN. ↑ AND ↓ REPRESENT HIGHER BETTER AND LOWER BETTER, RESPECTIVELY

Dataset	NIQE↓	BRISQUE↓	MUSIQ↑	DBCNN↑
Repaint	4.658	11.506	3.523	0.461
LaMa	4.476	10.546	3.476	0.463
Vaihingen	5.595	13.010	3.492	0.461

The results above indicate that the optical characteristics of the forgery regions in Fake-Vaihingen and Fake-LoveDA closely match those of authentic images, demonstrating high alignment in quality. In contrast, Splice-Vaihingen exhibits greater discrepancies in image quality compared to the generative model-produced Fake-Vaihingen dataset, underscoring the challenge posed by high-quality forgeries generated by advanced generative models for forgery localization and detection tasks.

Furthermore, we compare the generated image quality between Repaint and LaMa using both quantitative metrics and human perception, applying the same image sections and masks from the Fake-Vaihingen dataset. Inspection of Table IV reveals that Repaint's images are closer in quality to those of the authentic Vaihingen dataset on those metrics, presenting greater optical similarity. Visual comparisons of these generative models are presented in Fig. 1, illustrating that both Repaint and LaMa generate forgeries with fine texture details and overall visual perception.

IV. METHODOLOGY

A. Forgery Localization and Detection Collaborative Framework

1) Collaborative Framework: As depicted in Fig. 2, the proposed FLDCF integrates forgery localization and detection tasks to mutually enhance their performance. Forgery detection performance can vary across datasets due to differences in image content and forgery characteristics. The forgery masks generated by the forgery localization model provide crucial supplementary information that aids in the training of forgery detection models. By incorporating reliable reference mask data, we mitigate the performance instability of forgery detection across diverse landscapes, as well as improve overall detection accuracy. It is observed that even a simple forgery detection model endowed with accurate forgery mask information can achieve significant performance improvement across different datasets. Conversely, in forgery localization, detection

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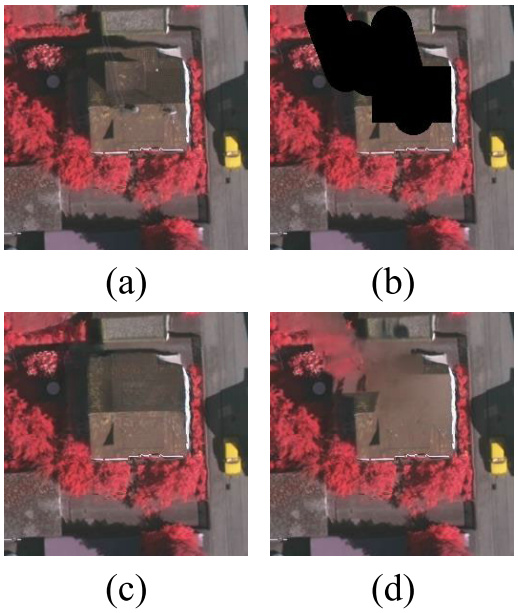


Fig. 1. Visual example of the forgery images in Fake-Vaihingen. (a) Authentic image. (b) Mask. (c) Forgery with LaMa. (d) Forgery with Repaint.

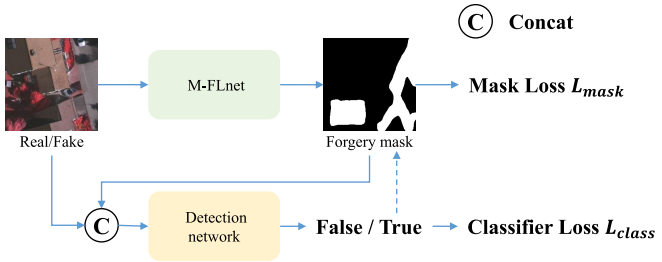


Fig. 2. Architecture of our FLDCF. The forgery mask generated by M-FLnet serves as an additional guidance for the training of the forgery detection network, while the results of the detection network correct the minor errors that appeared in the forgery mask of authentic images.

results help correct minor errors in the forgery masks. Specifically, with reference to the results derived from the forgery detection model, masks with minor forgery errors are ignored if the global image decision confirms their authenticity. As a result, this collaborative framework can enhance the accuracy of both forgery localization and detection tasks.

2) Localization and Detection Network: In our proposed collaborative framework, we develop a specialized network called M-FLnet to ensure precise localization, given the critical importance of accurate forgery localization for successful detection. Further details are provided in Section IV-B. Besides, as shown in Fig. 3, the forgery detection network serves a supplementary role, utilizing a lightweight ResNet structure with a two-class classifier designed to distinguish between two distinct classes.

B. M-FLnet

M-FLnet serves as the central module within our collaborative framework, which is responsible for generating forgery masks based on input images. Fig. 4 illustrates the architecture of M-FLnet, composed of two primary components: the main localization network and a content-based prior, which will be detailed in Section IV-C. The main localization network adopts an encoder–decoder structure with a ResNet backbone. The reduction of feature size conducted in the encoder and subsequent restoration in the decoder enhance the network's ability to perform integrated local–global learning. The encoder of M-FLnet captures high-level features from the input images, while the decoder reconstructs these features into the forgery mask, ensuring precise localization of manipulated areas.

1) Encoder: For the encoder in M-FLnet, the process begins with shallow convolutional layers that generate initial features. In addition, intermediate features from a content-based prior are integrated into the shallow feature extraction process. These combined features then undergo further processing through several residual blocks. These residual blocks are designed to facilitate deeper feature extraction by progressively reducing feature sizes, thereby enhancing the network's ability to learn deep local–global features effectively. It ensures that M-FLnet can capture and utilize hierarchical features crucial for precise forgery localization.

Toward the end of the encoder, a pyramid feature extraction module [50] is employed to further refine feature extraction across different dimensions. The structure of this module is depicted in Fig. 5. It comprises four parallel convolutional layers with varying kernel sizes and dilated rates, alongside a global average pooling (GAP) layer [51] with upsampling to enhance global feature extraction. Following the convolutional layers and GAP, all extracted features are concatenated and fused together to promote robust feature propagation and encourage feature reuse.

2) Decoder: Following the encoder in M-FLnet, a global decoder is employed to maintain the feedforward nature of the network. This decoder utilizes several refined upsampling modules to reconstruct the output. A refined upsampling module consists of two main components: an upsampling layer and a boundary refinement (BR) module [52]. The BR module, as illustrated in Fig. 6, employs a simple residual structure that passes through two convolutional layers with the ReLU activation between them. This design helps to refine boundaries and enhance the clarity of the generated mask.

3) Classifier Modules: In the final stage, M-FLnet produces a forgery mask that distinguishes between fake and authentic areas within the image using a two-class classifier. Traditionally, one-class classifiers that recognize instances of a single class and distinguish outliers or anomalies are preferred for forgery localization in remote sensing, focusing on identifying small abnormalities in images. Nowadays, advancements in generative models enable high-quality, large-area manipulations, making detection with a one-class classifier inadequate. Compared with one-class classifiers, two-class classifiers are designed to distinguish between two distinct classes, namely, the fake and authentic areas. As a result, the proposed M-FLnet is more suitable for handling various forgery scenarios encountered in contemporary satellite image analysis.

C. Content-Based Prior

Besides the aforementioned main localization network, a content-based prior is introduced to incorporate additional

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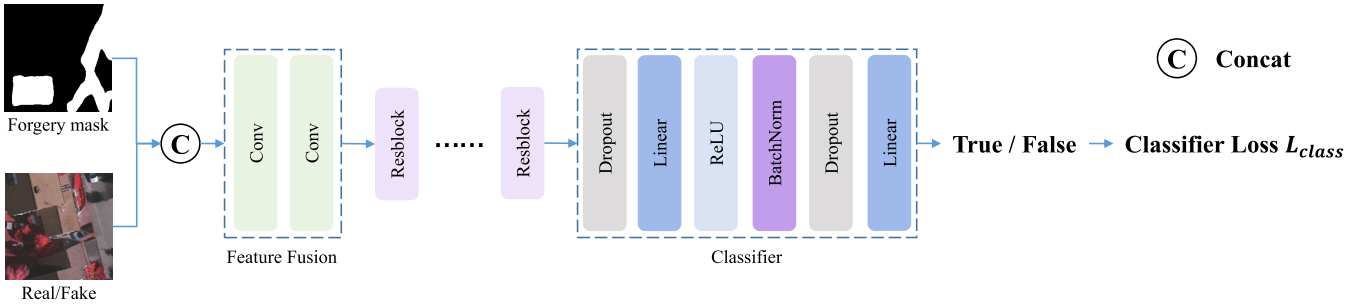


Fig. 3. Lightweight ResNet structure of the forgery detection network. It receives a fake or authentic image and the forgery mask generated by M-FLnet as inputs and identifies image authenticity on a global scale.

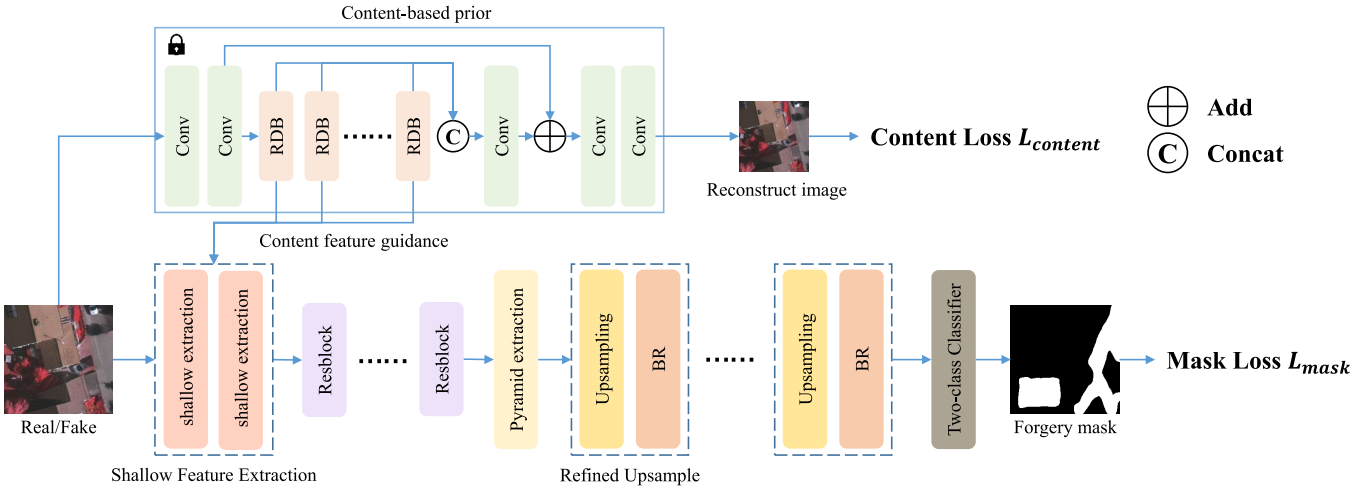


Fig. 4. Detailed architecture of M-FLnet. Pyramid extraction represents the pyramid feature extraction module. The content-based prior is trained separately beforehand and locked during the training of M-FLnet.

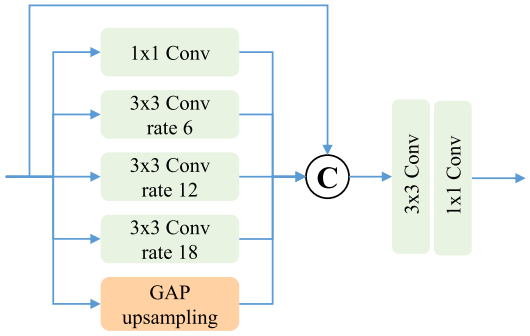


Fig. 5. Structure of the pyramid feature extraction module in M-FLnet.

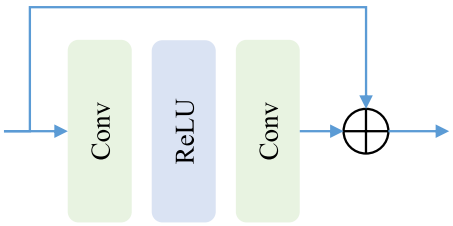


Fig. 6. Structure of the BR module. It contains two conventional layers and an ReLU activation function between them.

contextual information to refine the forgery mask generation process. The role of the content-based prior network is to restore the authentic image based on the fake image. During training, the network aims to preserve the landscape in authentic areas while predicting the original content of the forgery areas. This approach allows the network to learn global features of authentic scenes and specific features introduced by the generative model in the manipulated areas. To prevent confusion, we utilize features from the encoder as guidance since features in the decoder may contain more noise in postdecoding when producing less accurate results, potentially presenting inaccurate features to the localization network.

As depicted in Fig. 4, the structure of the content-based prior network follows the design of residual dense networks (RDNs) reported in [53]. Specifically, as shown in Fig. 7, within each residual dense block (RDB) [53] in RDN, features are divided into two groups: the first half and the second half. The features in each group are concatenated separately and passed through a convolutional layer to restore channels. Finally, the reconstructed features are concatenated with features from shallow feature extraction of the main localization network to enable effective feature fusion.

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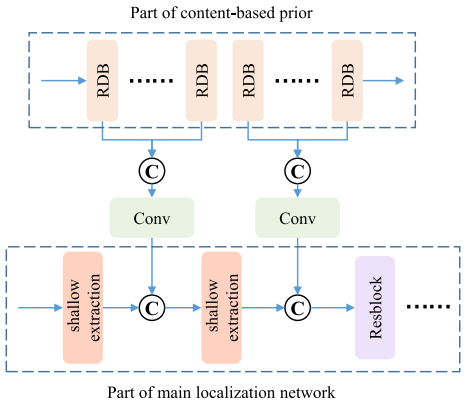


Fig. 7. Structure of the content feature guidance from content-based prior to the main localization network.

Given that the content-based prior network restores the entire image, it carries a higher computational complexity compared to the forgery localization task. To avoid computational burden on the forgery localization network, the content-based prior network is trained separately beforehand. Its parameters are fixed during the training of M-FLnet, ensuring stability and efficiency in the training process.

D. Loss Function

Three distinct loss functions, namely, the localization loss function, the classifier loss function, and the content loss function, are applied to train M-FLnet, forgery detection network, and content-based prior separately.

1) Localization Loss Function: For training M-FLnet to distinguish between fake and authentic areas in satellite images, the following cross-entropy loss function is applied:

$$L_{\text{mask}} = -\frac{1}{N}\sum_{n=1}^{N}\left[y_n \log(p_n) + (1-y_n)\log(q_n)\right] \tag{1}$$

where $N = W \times H$ with W and H being the width and height of the image. Furthermore, y_n is the ground truth, whereas p_n and q_n represent the possibility of this pixel being authentic or fake, respectively.

2) Classifier Loss Function: This loss function is designed for the forgery detection network, which aims to distinguish whether an image has undergone manipulation. It also utilizes the following cross-entropy loss function:

$$L_{\text{class}} = -y_{\text{all}} \log(p_{\text{real}}) - (1-y_{\text{all}})\log(p_{\text{fake}}) \tag{2}$$

where y_{all} represents the ground truth, while p_{real} and p_{fake} stand for the possibility of this image being authentic and fake, respectively.

3) Content Loss Function: The content loss function is implemented for the content-based prior network. It tries to restore the generated image back to the authentic image. The loss function minimizes the disparity between the restored and the authentic images, utilizing the following $L1$-norm loss:

$$L_{\text{content}} = \|I - I'\|_1 \tag{3}$$

where I and I' represent the authentic and restored images, respectively.

V. Experiments

A. Datasets and Metrics

To assess the effectiveness of our proposed method, we train and evaluate all forgery localization and detection models using our datasets: Fake-Vaihingen and Fake-LoveDA. The following performance metrics, namely, overall accuracy (OA), F1 score (F1), precision, and recall, are employed to assess the performance of forgery detection. The definitions for these metrics are listed in the following:

$$OA = \frac{\text{TP}+\text{FN}}{\text{TP}+\text{TN}+\text{FN}+\text{FP}} \tag{4}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{5}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{6}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{7}$$

where TP, FP, TN, and FN are true positives, false positives, true negatives, and false negatives, respectively. Furthermore, authentic images are considered true in the calculation of all metrics. As discussed in Table II, the number of manipulated images exceeds that of authentic images in both proposed datasets.

For the forgery localization, OA, mean F1 score (mF1), and mean Intersection over Union (mIoU) are defined as follows:

$$F1 = 2 \times \frac{P_c \times R_c}{R_c + P_c} \tag{8}$$

$$\text{IoU} = \frac{TP_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c} \tag{9}$$

where

$$P_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c} \tag{10}$$

$$R_c \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}. \tag{11}$$

Furthermore, TP_c, FP_c, TN_c, and FN_c are true positives, false positives, true negatives, and false negatives for each class, respectively. Note that mF1 and mIoU are calculated on the two classes, namely, authentic and forged areas.

B. Implement Detail

In the proposed FLDCF, the forgery detection network architecture adopts ResNet18, while the content-based prior network is constructed on RDN [53], featuring eight RDBs for image restoration. The number of feature channels is fixed at 32. The content-based prior network is trained independently and only evaluated during the training of M-FLnet and forgery detection. For M-FLnet, we employ four residual blocks and three refined upsampling modules. The batch size is configured as 16 for Fake-Vaihinge and 8 for Fake-LoveDA, reflecting their distinct image size requirements. All experiments are executed using PyTorch on a single NVIDIA GeForce RTX 4090 GPU equipped with 24 GB of RAM.

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TABLE V

QUANTITATIVE EXPERIMENTAL RESULTS (%) OF FORGERY LOCALIZATION ON THE FAKE-VAIHINGEN AND FAKE-LOVEDA DATASETS. THE BEST AND SECOND-BEST ARE BOLDED AND UNDERLINED, RESPECTIVELY

Method	Fake-Vaihingen			Fake-LoveDA		
	mIoU(%)	OA(%)	mF1(%)	mIoU(%)	OA(%)	mF1(%)
SCSE-Unet [38]	82.78	88.93	90.18	97.68	98.76	98.82
MVSSNet [24]	81.62	87.07	89.40	97.15	98.57	98.54
SE-Network [17]	<u>84.45</u>	<u>90.23</u>	<u>91.25</u>	<u>98.37</u>	<u>99.21</u>	**99.28**
Proposed	**90.64**	**94.79**	**94.98**	**98.50**	**99.23**	<u>99.24</u>

TABLE VI

QUANTITATIVE EXPERIMENTAL RESULTS (%) OF FORGERY DETECTION ON THE FAKE-VAIHINGEN AND FAKE-LOVEDA DATASETS. THE BEST AND SECOND-BEST ARE BOLDED AND UNDERLINED, RESPECTIVELY

Method	Fake-Vaihingen				Fake-LoveDA			
	Precision(%)	Recall(%)	OA(%)	F1(%)	Precision(%)	Recall(%)	OA(%)	F1(%)
Capsule [39]	90.48	82.01	92.95	86.04	79.75	75.47	89.13	77.55
FaceForensics [25]	91.03	94.96	96.19	92.96	<u>99.53</u>	**100**	<u>99.88</u>	<u>99.77</u>
DeepFake-Detection [7]	95.20	85.61	95.05	90.15	87.53	96.73	95.76	91.90
Geo-DefakeHop [6]	<u>98.68</u>	<u>97.15</u>	<u>96.95</u>	<u>97.91</u>	95.12	94.86	92.80	95.18
Proposed	**99.29**	**100**	**99.81**	**99.64**	**100**	**100**	**100**	**100**

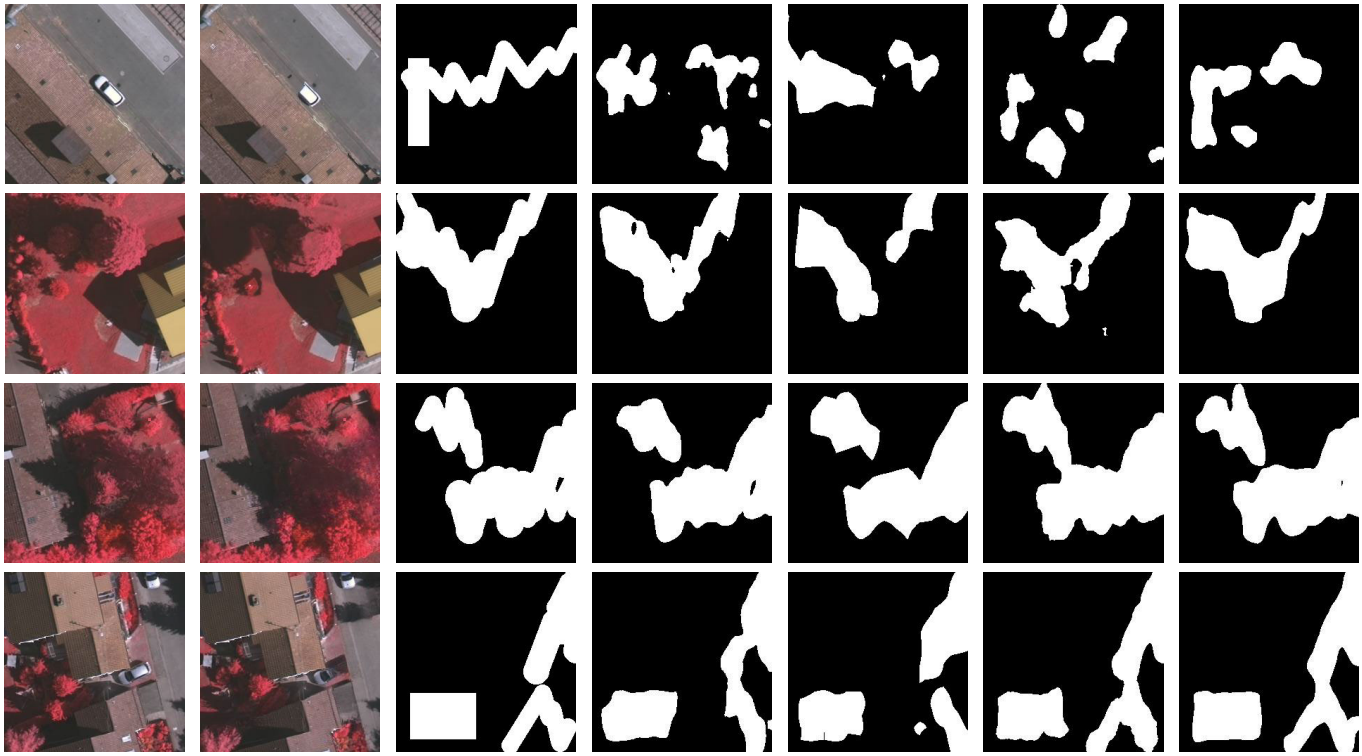


Fig. 8. Visual comparisons on Fake-Vaihingen. (a) Authentic images. (b) Fake images (the first two rows stand for the forged images generated with repaint where the last two rows are with LaMa.) (c) Ground truth. (d) SCSE-Unet. (e) MVSSNet. (f) SE-Network. (g) Proposed.

C. Performance Comparison

To demonstrate the effectiveness of our FLDCF, the following state-of-the-art forgery detection models, namely, Capsule [39], FaceForensics [25], DeepFake-Detection [7], and Geo-DefakeHop [6], and three forgery localization models, including SCSE-Unet [38], MVSSNet [24], and SE-Network [17], are evaluated for comparison. The quantitative results of forgery localization and detection on the Fake-Vaihingen and Fake-LoveDA are presented in Tables V and VI.

1) Localization: As shown in Table V, our method outperforms other forgery localization models across all datasets. Particularly in Fake-Vaihingen, our model demonstrates significant improvements of 6%, 4%, and 3% in mIoU, OA, and mF1, respectively. These gains across all metrics indicate our model's superior ability to distinguish between authentic and fake locations across diverse landscapes. Models designed for traditional forgery localization, such as SE-Network, exhibit notably lower accuracy compared to our M-FLnet, highlighting their limitations in handling high-quality manipulations. In addition, as depicted in Table VII, our model consistently

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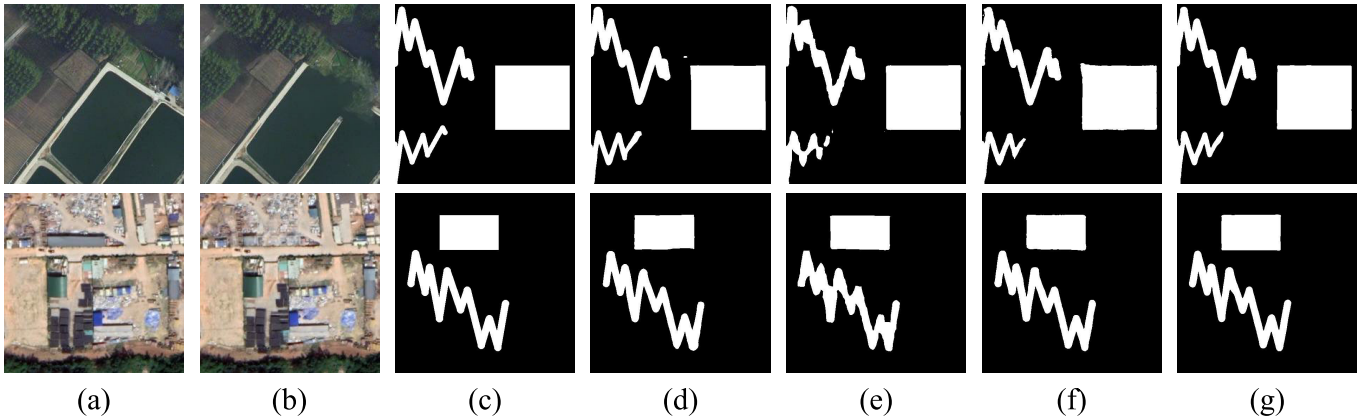


Fig. 9. Visual comparisons on Fake-LoveDA. (a) Authentic images. (b) Fake images (the first row stands for the forged images generated with repaint where the last row is with LaMa.) (c) Ground truth. (d) SCSE-Unet. (e) MVSSNet. (f) SE-Network. (g) Proposed.

outperforms other localization models across both classes in terms of IoU and F1. Furthermore, an inspection of Table V suggests that Fake-LoveDA is less challenging compared to Fake-Vaihingen, as evidenced by the much higher forgery localization performance scores. Both generative models are trained on natural images with detailed texture generation capabilities. However, for datasets such as LoveDA, which have low resolution and less detailed information, the detailed generation results in lower quality forgeries that tend to display more obvious spectral characteristics. It is easier to detect for both conventional models and our proposed model. As a result, all models under investigation demonstrate similar forgery localization performance on Fake-LoveDA. Despite that the proposed model scores 0.04% lower in mF1 compared to the top-performing SE-Network, it outperforms other models in the remaining two metrics, mIoU and OA, on Fake-LoveDA and achieves significantly better results across all three metrics on the more challenging Fake-Vaihingen dataset.

Visual results for Fake-Vaihingen and Fake-LoveDA are presented in Figs. 8 and 9, respectively. For Fake-Vaihingen, models such as SCSE-Unet, MVSSNet, and SE-Network struggle with inaccurate predictions and omissions, whereas our proposed method achieves superior performance with precise mask outlines. For instance, our model accurately outlines the fake areas in the second image generated by LaMa, while other localization models fail. Furthermore, Fig. 9 shows that all models perform well on the less challenging Fake-LoveDA and generate very similar masks. Nevertheless, our model excels by providing superior results.

2) Detection: Table VI shows the forgery detection performance. As shown, the proposed forgery detection model, enhanced with forgery mask guidance, demonstrates superior performance compared to the other forgery detection models. More specifically, our model achieves improvements of 0.6%, 2.9%, 2.9%, and 1.7% in precision, recall, OA, and F1 scores, respectively, compared to the second-best model Geo-DefakeHop. Furthermore, the use of mask guidance ensures that our detection results are less susceptible to variations in image content or forgery quality. In sharp contrast to existing models that are affected by variations in image content and forgery quality, the proposed FLDCF consistently achieves near-perfect scores across different datasets, approaching 100% in each metric.

TABLE VII

QUANTITATIVE EXPERIMENTAL RESULTS (%) ON AUTHENTIC AND FAKE CLASS ON FAKE-VAIHINGEN DATASET. THE BEST IS GIVEN BOLDFACE

Method	Authentic		Fake	
	IoU	F1	IoU	F1
SCSE-Unet [38]	91.02	95.30	71.84	83.61
MVSSNet [24]	90.70	95.12	69.73	82.16
SE-Network [17]	92.01	95.84	74.85	85.61
Proposed	**95.12**	**97.50**	**84.62**	**91.67**

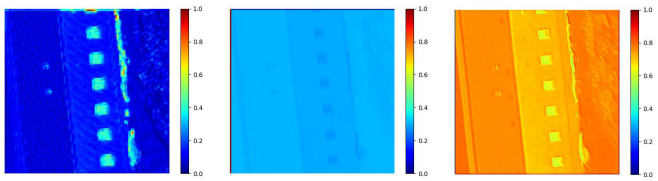


Fig. 10. Visual inspection of the activation maps of (a) features from the encoder in the main localization network, (b) reconstructed features from the content-based prior, and (c) features from the encoder of the content-based prior network.

D. Content-Based Prior Analysis

Fig. 10 compares the activation maps of features derived from the shallow feature extraction of the main localization network, reconstructed features from the content-based prior after the concatenation and convolutional layer, and features from the encoder of the content-based prior network. As depicted in Fig. 10(a), the shallow feature extraction of the main localization network focuses primarily on local information, highlighting specific areas with high values. Fig. 10(b) shows the reconstructed features from the content-based prior. Finally, an inspection of Fig. 10(c) suggests that a greater emphasis is placed on global information for the encoder of the content-based prior network. As discussed in Section III-B, the content-based prior network aims to restore the authentic

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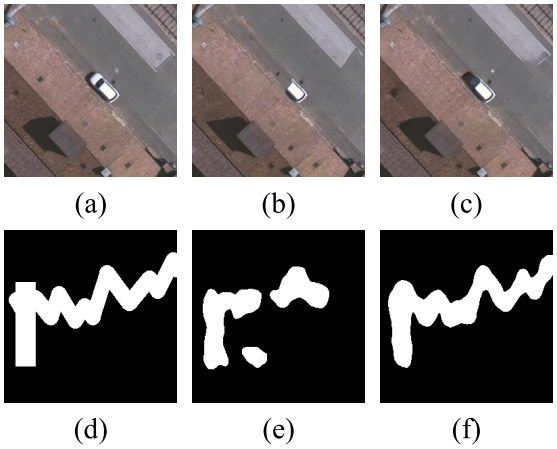


Fig. 11. Visual inspection of the authentic image and generated image based on Repaint and LaMa with the same mask and their localization result. (a) Authentic image. (b) Image generated by Repaint. (c) Image generated by LaMa. (d) Ground-truth mask. (e) Localization result on image generated by Repaint. (f) Localization result on the image generated by LaMa.

TABLE VIII
LOCALIZATION PERFORMANCE (%) ON THE FAKE-VAIHINGEN DATASET CREATED BY DIFFERENT GENERATIVE MODELS

Method	Fake-Vaihingen		
	mIoU	OA	mF1
Repaint	85.64	91.58	92.09
LaMa	93.36	96.83	96.53

image based on the fake images, thereby learning specific features of manipulation and general features of authentic areas. The notable disparity between these activation maps underscores the distinct objectives of the content-based prior and the main localization network, demonstrating that the content-based prior indeed provides additional valuable information for training the forgery localization model.

E. Generative Model Analysis

Table VIII and Fig. 11 present experimental results on the forgery localization performance using forgery created by two different generative models. More specifically, the models are trained on the Fake-Vaihingen training set and tested on forgery images generated separately by Repaint and LaMa. As depicted in Table VIII and Fig. 11, the localization performance on LaMa-generated images is notably better than on those generated by Repaint, indicating that it is easier to detect forgery locations generated by LaMa.

F. Mask Analysis

Forgery localization and forgery generation represent adversarial applications. To analyze the influence of manipulated masks on forgery localization, we conduct experiments varying the shape and area of the manipulated masks. These experiments focus solely on understanding how the mask characteristics affect localization results, without considering specific image content. Unless stated otherwise, the proposed M-FLnet is utilized in all subsequent experiments.

TABLE IX
LOCALIZATION PERFORMANCE (%) WITH IRREGULAR AND REGULAR FORGERY MASKS ON THE FAKE-VAIHINGEN DATASET

Method	Fake-Vaihingen			
	mIoU	OA	mF1	Mask
Irregular	93.54	96.69	96.65	30.14
Regular	96.78	98.30	98.36	29.79

TABLE X
QUANTITATIVE EXPERIMENTAL RESULTS (%) OF FORGERY LOCALIZATION BASED ON FORGERY MASK WITH DIFFERENT AREAS ON THE FAKE-VAIHINGEN DATASET

Area	Fake-Vaihingen		
	mIoU	OA	mF1
10%	94.80	97.31	97.28
20%	96.22	97.88	98.06
30%	96.87	98.30	98.41
40%	96.47	98.09	98.20

1) Forgery Shape: To evaluate the impact of forgery mask shapes on localization performance, we conduct experiments on the Fake-Vaihingen test set with forgery generated by LaMa using either irregular or regular masks. In particular, the irregular masks are generated in a casual painting style, while regular masks are uniformly squares and randomly placed within images, each occupying approximately 30% of the image area. As shown in Table IX, irregular forgery locations prove more challenging for detection, which is evidenced by the poorer localization performance.

This phenomenon can be attributed to factors related to both generative quality and localization capability. In terms of generative quality, irregular masks facilitate higher quality forgery placements because generative models can utilize adjacent pixels more effectively as references. This allows the model to better imitate image features and seamlessly fill irregularly shaped masked areas. Conversely, regular masks, especially the center area of squares, provide less reference for the generative model, resulting in noticeable discrepancies between authentic and fake areas. Consequently, the high-quality generation associated with irregular masks increases the difficulty of the forgery localization.

From a localization perspective, it is easier for the network to outline forgery locations with clear, regular edges compared to irregular outlines. Given that both irregular and regular masks are present in the training data, the model learns to recognize and outline both types of shapes. The uniformity of regular forgery masks in the test set aligns with those seen during training, further enhancing the model's performance on simpler, regular mask shapes.

2) Forgery Area: In addition to the forgery shape, the mask size also significantly influences the forgery localization performance. Table X presents results from testing on various mask sizes, ranging from 10% to 40%, using square masks centered in the image and generated by the same model, i.e., LaMa.

As shown in Table X, the localization performance first improves as the mask size increases from 10% to 30%, peaking at 30%. For masks covering 10%–30% of the image area, the generative model produces worse forgery replacements by

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TABLE XI

QUANTITATIVE EXPERIMENTAL RESULTS (%) OF FORGERY LOCALIZATION ON SPLICE-VAIHINGEN

Method	Splice-Vaihingen		
	mIoU	OA	mF1
SCSE-Unet	97.14	98.22	98.54
MVSSNet	94.09	96.62	96.92
SE-Network	84.96	89.51	91.57
Proposed	96.71	97.96	98.32

TABLE XII

ABLATION STUDY OF FORGERY LOCALIZATION ON THE FAKE-VAIHINGEN FOR CONTENT-BASED PRIOR

Content-based prior	mIoU	OA	mF1
	90.01	94.46	94.62
✓	90.64	94.79	94.98

TABLE XIII

ABLATION STUDY OF FORGERY DETECTION ON THE FAKE-VAIHINGEN FOR MASK GUIDANCE

Mask guidance	Precision	Recall	OA	F1
	97.39	80.58	94.29	88.19
✓	99.64	99.81	99.29	100

utilizing less reference information, making it less challenging for the localization model to detect forgery features. However, the localization accuracy does not further improve beyond 30% due to the limited authentic areas in the image available for reference, making it increasingly difficult to accurately localize forgeries beyond the 30% mask area threshold.

G. Localization Performance for Conventional Manipulation

To illustrate the effectiveness of the proposed framework and datasets, we construct an additional forgery dataset named Splice-Vaihingen with conventional manipulations for further experiments. As shown in Table XI, all models achieve high performance on this dataset generated with conventional manipulations, which is in sharp contrast to their performance in Fake-Vaihingen reported in Table V. This experiment highlights the necessity for addressing the challenges posed by improved high-quality forgery generation scenarios.

H. Ablation Study

Tables XII and XIII present the results of an ablation study investigating the impact of content-based prior for forgery localization and mask guidance for forgery detection. In Table XII, the inclusion of content-based prior results in improvements of 0.6%, 0.3%, and 0.4% in mIoU, OA, and mF1 for forgery localization, respectively. Meanwhile, Table XIII demonstrates that mask guidance leads to significant improvements of 2%, 19%, 5%, and 12% in the forgery detection task of the collaborative framework.

In addition, Fig. 12 depicts the OA convergence behavior over iterations. Despite initial fluctuations caused by unstable mask guidance in the early stages, the forgery detection achieves high final accuracy after the convergence of both the forgery localization and detection processes. This ablation study underscores the advantages of our FLDCF and the proposed content-based prior.

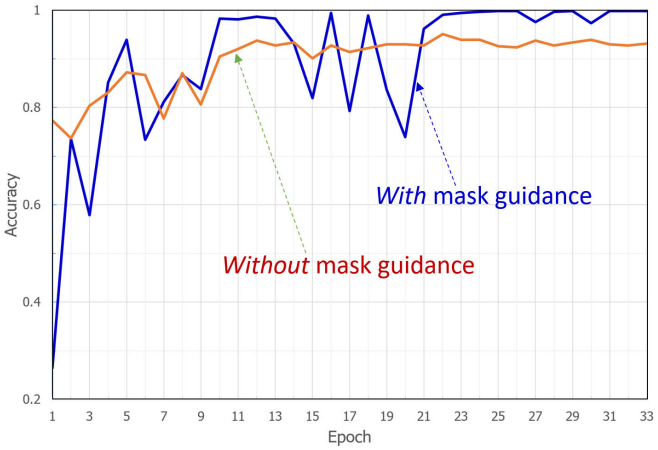


Fig. 12. Trend of OA testing in each training iteration on Fake-Vaihingen with or without the mask guidance.

TABLE XIV

COMPUTATIONAL COMPLEXITY OF THE COMPARED METHODS. PRIOR MEANS OUR CONTENT-BASED PRIOR NETWORK

Models	Complexity (G)	Memory (MB)	Parameters (M)	Speed (FPS)
SCSE-Unet	288.67	9436	41.25	16.05
MVSSNet	1423.70	11044	146.88	1.03
SE-Network	157.05	10102	3.94	14.92
Prior	297.09	20212	0.28	16.99
Proposed	752.26	11522	31.27	7.72

I. Computational Complexity Analysis

Finally, we conduct a comprehensive comparison of computational complexity across various models, focusing on model complexity, memory usage, parameter count, and processing speed. As detailed in Table XIV, the findings indicate that our content-based prior is a lightweight network. The substantial memory comes from its unique training target. Unlike other localization networks that only produce a boolean mask to indicate forgery locations, this model focuses on reconstructing the original image from the altered version. Regarding our proposed M-FLnet, it achieves superior results without substantially increasing computational complexity.

VI. CONCLUSION

In this work, an FLDCF has been proposed for satellite imagery. The forgery mask produced by the localization network provides additional guidance for training the detection network. In turn, the detection network's decisions correct minor errors in the forgery mask for authentic images. In particular, this work has developed M-FLnet to extract the image features based on multiple views to provide precise outlines of forgery areas. Furthermore, the proposed M-FLnet leverages a content-based prior to incorporate additional feature information, enhancing forgery localization accuracy. To evaluate the collaborative framework's performance in forgery localization and detection, two novel generative satellite forgery datasets, namely, Fake-Vaihingen and Fake-LoveDA, have been developed. In sharp contrast to existing forgery localization and detection datasets, these two new datasets are featured with

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high-quality forgery images created by two advanced generative models, namely, Repaint and LaMa, which represent a significant advancement over traditional forgery datasets used in remote sensing. Our experimental results on Fake-Vaihingen and Fake-LoveDA datasets have confirmed the superior performance of the proposed FLDCF.

There are several promising extensions to this study. First, our current datasets are generated using randomly created masks without specific guidelines. An interesting next step would be to explore masks designed for specific objects, such as cars and buildings, to simulate realistic forgery scenarios better. Finally, further performance improvements could be achieved by developing more effective feature fusion techniques to better integrate content-based priors with the main localization network in M-FLnet.

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